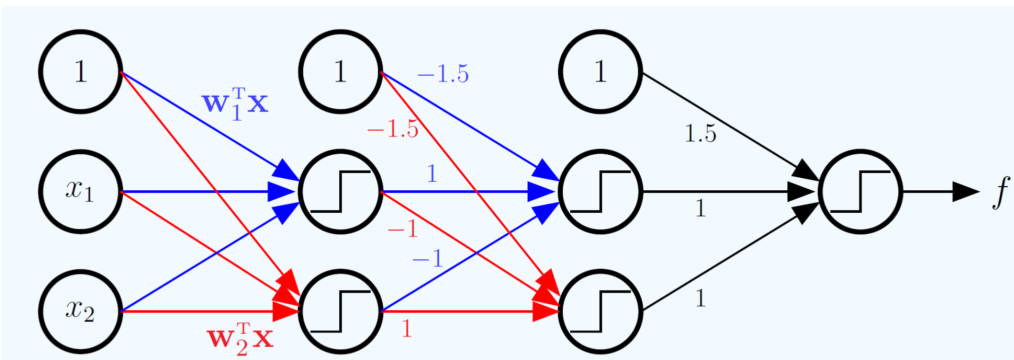
# HW -- Neural Networks

Understanding of neural networks and implementing one from scratch

Due: Thursday, October 30th, 11:59pm Pacific Time

## Task 1

Use the below graph representation to compute an explicit formula for f in terms of $\vec{x} = [1\ x_1\ x_2]^T$




## Task 2

Part 1: Write a function to perform forward propagation and back propagation to compute the gradient on a neural network with 2 layers. Your neural network should have 2 input units ($d^{(0)} = 2$), m hidden units ($d^{(1)} = m$), 1 output unit ($d^{(2)} = 1$). Use tanh as the activation function on the neurons. For gradient, use squared error as the error measure.

If you would like to make the code generic, think about how you need to specify the neural network with the layers and the number of neurons per layer. E.g., the functions for forward and back propagation may take an array of $W^{(l)}$ matrices and the input vector $x^{(0)}$ as the parameters.

Part 2: Implement gradient descent to train the neural network you implemented. Make your code capable of accepting any number of samples, so that you can easily perform stochastic gradient descent using this function.

## Submission instructions

We will use GitHub to share code and track progress in this class. If you have not already done so, create a GitHub private repository for the course and name it "CMPE257-Fall25-FirstName-LastName". Add the following users to the repository: Arielle00. You are also welcome to add the instructor and ISA to your Colab notebooks. Please include a README on how to run your code.

Include a link to the GitHub repository in your submission.